

ARASH VASHAGH

✉ Email — 📞 +1 (506)-238-7170 — 🔗 LinkedIn — 🎓 Google Scholar — 🌐 Portfolio — 📍 Fredericton, NB

Summary

Machine Learning Developer with research publications and industry certifications in AI, cybersecurity, and cloud. Experienced in adversarially resilient models and end-to-end deployed AI applications using PyTorch, Streamlit, FastAPI, and cloud platforms such as AWS and Azure. Strong at translating research into practical products for users.

Skills

Programming: Python, Java, JavaFX, Kotlin, SQL, C++

AI/ML: PyTorch, TensorFlow, Scikit-learn, Adversarial Machine Learning, RAG/LLMOps

Web, Backend & APIs: Streamlit, FastAPI, Flask, REST APIs, OpenAI API, JSON processing

Cloud & Deployment: AWS, Microsoft Azure, Streamlit Community Cloud, Docker, GitHub

Tools & Libraries: Jupyter, VS Code, IntelliJ IDEA, Git, MLflow, Wireshark, pypdf, ReportLab

Languages: Persian (Native), English (Professional)

Certifications

- **AI and Machine Learning:** AWS Certified AI Practitioner; Microsoft AI-900; DeepLearning.AI Agentic AI
- **Cloud Computing:** AWS Certified Cloud Practitioner; Microsoft AZ-900; Google Cloud Digital Leader (planned)
- **Cybersecurity:** CompTIA Security+; Microsoft SC-900
- **Data and Project Management:** CompTIA Data+; Microsoft DP-900; CompTIA Project Management Essentials; CompTIA DataSys+ (results pending)

Experience

Graduate Research Assistant (Team Lead) — UNB — Fredericton, Canada Sep 2024 – Present

- Conducting MSc thesis research in Adversarial Machine Learning (AML), focusing on adversarial attacks, defense mechanisms, and trustworthy AI systems.
- Co-authored [a survey](#) on adversarial attacks across machine learning and explainable AI.
- Co-authored a survey on defense mechanisms in adversarial machine learning.
- Proposed three adversarial defense mechanisms, with implementation and manuscripts in progress.
- Developing a fusion-model adversarial attack, with implementation and manuscript in progress.
- Coordinating graduate student work and managing the [TSAI LinkedIn page](#) for lab updates and educational content.

Graduate Teaching Assistant — UNB — Fredericton, Canada Jan 2025 – Apr 2026

- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Conference Reviewer — Remote Jul 2024 – Oct 2024

- Reviewed manuscripts and provided structured feedback at international conferences:
 - *ArtInHCI 2024* – Artificial Intelligence and Human-Computer Interaction (Kunming, China) [\[Certificate\]](#)
 - *BIBE 2024* – Biological Information and Biomedical Engineering (Hohhot, China) [\[Certificate\]](#)

Research Assistant — Isfahan University of Technology (IUT) — Isfahan, Iran Feb 2021 – Feb 2024

- Conducted research on machine learning and healthcare, resulting in two IEEE publications [\[1\]](#) and [\[2\]](#).
- Collaborated with faculty on model design, data preprocessing, and experimental evaluation.
- Supported technical writing and presentation preparation for conferences.

Teaching Assistant — IUT — Isfahan, Iran Feb 2021 – Feb 2024

- Courses: Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C++, Java, and Python.
- Gave tutorials and graded assignments.

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada Sep 2024 – Aug 2026

- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran Sep 2019 – Feb 2024

- Major: Computer Engineering
- Minor: Intelligent Systems

Selected Projects

PaperWise AI: Deployed Research Paper RAG / LLMOps App

Jun 2026

- Built and deployed a public Streamlit app for research paper question answering.
- Implemented PDF ingestion, text extraction, chunking, TF-IDF retrieval, OpenAI API integration, and offline fallback.
- Added structured paper summaries, suggested questions, reviewer critique mode, multi-paper comparison, and PDF answer export.
- Displayed RAG pipeline metrics including retrieval scores, response latency, retrieved chunks, and citation usage.
- Technologies: Python, Streamlit, Scikit-learn, OpenAI API, pypdf, ReportLab, Streamlit Community Cloud.
- Live app: paperwise-ai.streamlit.app.
- Source code: [GitHub](#).

PromptShield: Adversarial Detection for Vision-Language Models

Jan 2026 – May 2026

- Developed a machine learning framework to detect adversarial images in vision-language models using representation-level features.
- Evaluated detection performance across multiple datasets, model backbones, and adversarial attack settings.
- Designed experiments for robustness evaluation, attack transferability analysis, and detector performance comparison.
- Built reproducible ML experiments for model training, evaluation, metric tracking, and result analysis.
- Technologies: Python, PyTorch, OpenCLIP, NumPy, Scikit-learn, adversarial machine learning.
- Manuscript and code under review at [NeurIPS 2026](#).
- Source code: [GitHub](#).

AttackBench: Adversarial Attack Visualization Tool

Mar 2026 – Apr 2026

- Developed a desktop application to generate, run, and visualize adversarial attacks on image classification models.
- Implemented a JavaFX user interface with a Python FastAPI backend for real-time model inference and attack execution.
- Displayed model predictions, confidence scores, adversarial examples, and attack impact for user interpretation.
- Integrated REST API communication between frontend and backend components.
- Technologies: Java, JavaFX, Python, PyTorch, FastAPI, REST APIs, adversarial machine learning.
- Source code: [GitHub](#).

Voluntary Educator (AI Security)

AI Security and Adversarial Machine Learning Course — Online

Jun 2026 – Present

- Creating a Persian technical course on AI Security and Adversarial Machine Learning to support Persian-speaking learners facing language barriers.
- Developed slides and scripts on machine learning foundations for AML, including classification, loss functions, gradients, and gradient descent.
- Published course videos through a structured [YouTube playlist](#) and shared course materials on [GitHub](#).
- Targeting completion of the Persian course by the end of June 2026, with an English version planned afterward.